

ELEC50001 – Circuits and Systems

SAMPLE PAPER 1

SOLUTIONS

Answer ALL questions.

There are THREE questions on the paper.

Question ONE counts for 50% of the marks, other questions 25% each

Time allowed: 2 hours

1. (a) Figure 1.1 shows an amplifier circuit using an op-amp operating at 3.3V and has a gain-bandwidth product of 1MHz. It amplifies the input signal V_S and produces an output voltage V_{out} .

(i) Derive from first principle the gain of this amplifier for both the DC and AC components of V_S . You must show your derivation clearly. [4]

(ii) Given that V_S is a sine wave signal at 20kHz with a peak-to-peak voltage of 1V, sketch and explain the expected waveform at V_{out} . [3]

(iii) State with justifications what happens to the gain of the amplifier if the frequency of V_S is changed to 1Hz and to 1MHz. [3]

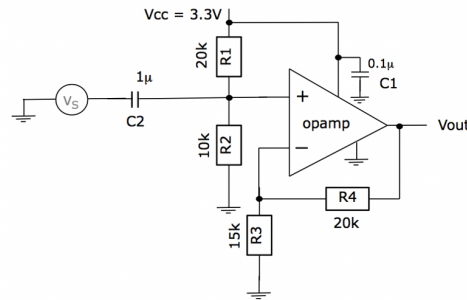


Figure 1.1

SOLUTION

This question examines student's ability to use an op-amp and demonstrate their understand of its limitations.

(i) (Bookwork, but from first principle). C_2 blocks dc from V_S . Therefore the gain of the dc offset of V_S is 0. R_3 and R_4 forms a voltage divider and must be equal to ac signal of V_S .

$$V_{out} * \frac{R_3}{R_4 + R_3} = V_S$$

Hence the gain is $(1 + R_4/R_3) = 2.33$.

Also, student may derive the equation relating V_{out} to V_S :

$$V_{out} = \frac{R_4 + R_3}{R_4} * \left(V_S + 3.3 * \frac{R_2}{R_1 + R_2} \right) = 2.33 * V_S + 2.56$$

[4]

(ii) V_+ dc voltage is determined by R_1 and R_2 and it is 1.1V. The gain of the amplifier is 2.33 for both dc and ac at V_+ . Therefore V_{out} an ac signal with pk-pk voltage of $2.33 \times 1V = 2.33V$, with a dc offset of $2.33 \times 1.1V = 2.56V$.

Therefore $V_{out(max)} = 3.73V$ and $V_{out(min)} = 1.4V$. Unfortunately the power supply is at 3.3V. Therefore V_{out} is clipped for any voltage above 3.3V.



[3]

(iii) R_1 and R_2 gives a total resistance of $R_1 \parallel R_2 = 6.67k$, with C_2 , the corner frequency is 24Hz. Therefore 1Hz sine wave does not pass through C_2 unimpeded. Since the GBP is 1MHz, it also cannot provide a gain of 2.33 at this signal frequency. Therefore the gain of the amplifier is lower than the predicted 2.33.

[3]

- (b) (i) Explain the meaning of the following specifications for a digital-to-analogue converter (DAC): resolution, linearity error, monotonicity and settling time. Give one example where the monotonic behaviour of a DAC is important. [4]

- (ii) Figure 1.2 shows a circuit for a R-2R 4-bit DAC with input $IN[3:0] = 4'b0110$. The resistor network is supplied via a $8mA$ ideal current source. The digital inputs $IN[3:0]$ control the four 2-way electronic switches. The output V_{out} is driven by an ideal operational amplifier. [4]

Derive the values for the current I_o , and voltages V_0 , V_2 and V_{out} .

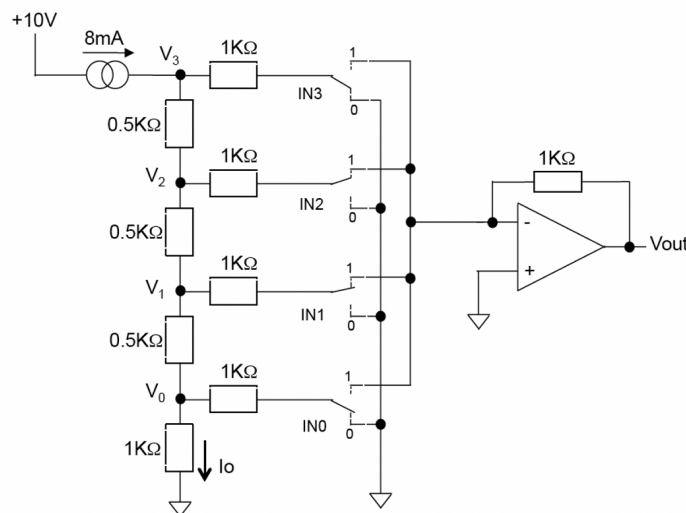


Figure 1.2

Q1 (b) This question tests students understanding of R-2R DAC converters.

- (i) This part of the question is purely bookwork.

Resolution – the voltage step equivalent to one least significant bit change of the digital number. Assuming that the digital number is N-bits, then this is the same as Full-scale voltage / $(2^N - 1)$.

Linearity error – this is the maximum deviation of the output vs input characteristic from the linear line joining the maximum point with the minimum point.

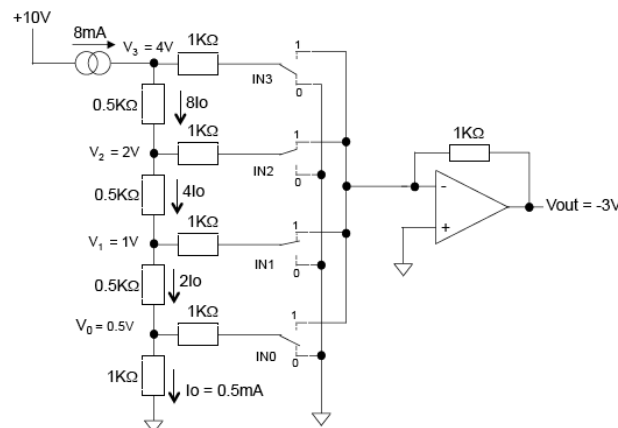
Monotonic DAC – One that always goes up as the input number increases.

Settling time – Time taken to reach final value as input changes.

Monotonic behaviour is most important in a control loop implementation using DAC. If the characteristic is not monotonic, the gain of the system may change sign, and could then cause the system to become unstable.

[4]

- (ii) The currents and voltages are shown below.



[4]

(c) Figure 1.3 shows a 4k x 4 RAM chip with address bus ADR, data bus DATA, a write enable signal WR, a chip select signal CS and a synchronising clock CLK.

(i) Specify the size of the address and data buses for the RAM chip. [2]

(ii) An 8k x 8 bit RAM module is required in a microprocessor system with a 16-bit address bus and an 8-bit data bus. The starting address of the RAM module is 16'h8000.

a) Design, in the form of a schematic diagram, a circuit using four 4k x 4 RAM chips shown in Figure 1.3 to implement the 8k x 8 RAM module. [5]

b) Derive the Boolean expression for the chip select signals required for each of the 4k x 4 RAM chip. [3]

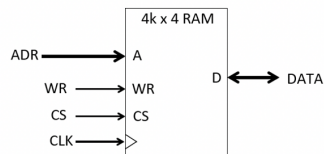
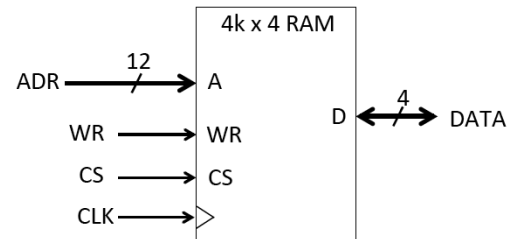


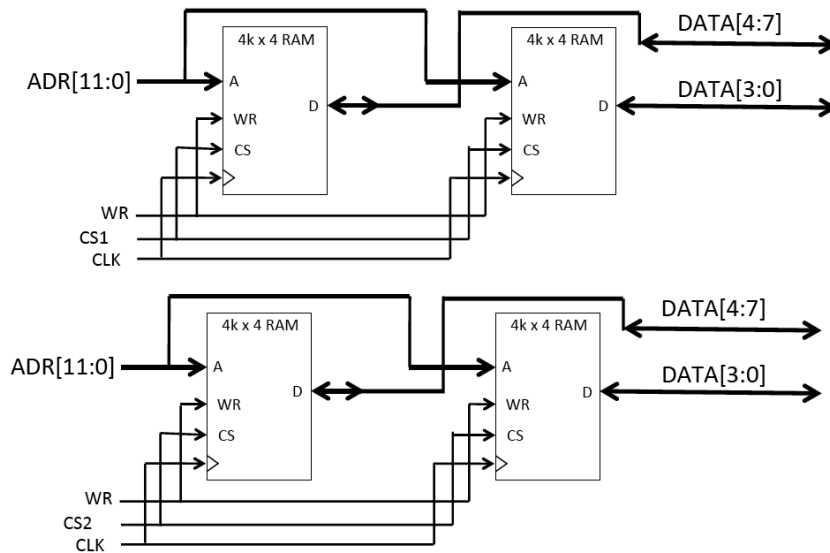
Figure 1.3

Q1 c) This question tests student's understanding of memory organisation and address decoding.



(i) Address: 12 bits, Data 4 bits.. [2]

(ii) a)



[5]

b) CS1 address range: 16'h8000 to 16'h8fff.

CS2 address range: 16'h9000 to 16'h9fff.

Therefore:

$$CS1 = A_{15} \& !A_{14} \& !A_{13} \& !A_{12}$$

$$CS2 = A_{15} \& !A_{14} \& !A_{13} \& A_{12}$$

[3]

- (d) Consider the circuit shown in *Figure 1.4*. The propagation delay t_d of the inverters G1 to G3, and the clock-to-Q delay t_{cq} of the flip-flops FF1 and FF2, are in the range of 0.5ns to 1.0ns. The setup and hold times of the flip-flops FF1 and FF2 are 1.3ns and 1.1ns respectively. The clock to FF2 is driven through a delay element DLY with a delay of t_{dly} .
- (i) Assuming that $t_{dly} = 0$ ns and using only the setup time constraint, calculate the maximum operating frequency of the clock signal CLK. [4]
- (ii) Assuming that $t_{dly} = 0$ ns, show that there is a hold time violation for FF1. [3]
- (iii) Delaying the clock to FF2 can eliminate the hold time violation. Show that the hold time violation is eliminated if the delay $t_{dly} > 0.1$ ns. [3]

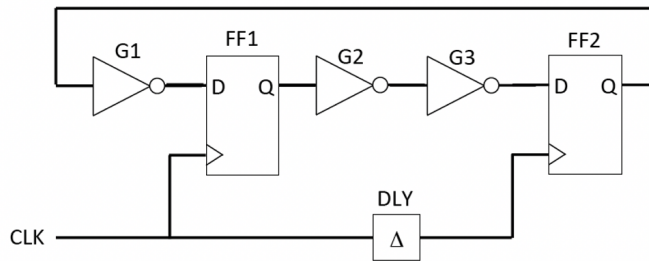


Figure 1.4

Q1 (d) This question tests student's understanding of the timing constraints in digital circuits.

(i)

Ignoring any hold time violation, the path from Q1 (FF1 Q output) to D2 governs the worst-case timing. This gives:

$$t_{clk} \geq 2 * t_{d(max)} + t_{cq(max)} + t_{su}$$

Therefore $t_{clk} \geq 2 * 1\text{ns} + 1\text{ns} + 1.3\text{ns} = 4.3\text{ns}$

Hence $f_{clk(max)} = 232.6\text{MHz}$.

[4]

(ii)

The shortest delay from Q2 (FF2 Q output) to D1 (FF1 D input) is:

$$t_{cq(min)} + t_{d(min)} = 1.0 \text{ ns}$$

which is shorter than the hold time requirement of FF1, i.e. 1.1ns. Therefore there is hold time violation at FF1 D input.

[3]

(iii)

To eliminate the hold time violation, we want to delay the arrival time at D1 by at least 0.1ns. This can be achieved by delaying the clock to FF2 by the same amount of 0.1ns. Therefore the minimum value required for t_{dly} is 0.1ns.

[3]

(e) Figure 1.5a shows the Verilog specification of a finite state machine FSM with four states that uses one-hot encoding.

(i) What is one-hot encoding? Briefly explain why one-hot encoding is particularly suitable for FPGA implementation. [4]

(ii) Construct the state transition diagram for this FSM. [4]

(iii) Hence or otherwise, complete the timing diagram shown in Figure 1.5b. [4]

```

module FSM (x, a, clk, rst);
input  a, clk, rst;
output x;

// define states - one-hot encoding
parameter NSTATE = 4;
parameter S_0 = 4'b0001;
parameter S_1 = 4'b0010;
parameter S_2 = 4'b0100;
parameter S_3 = 4'b1000;

reg [NSTATE-1:0] state;
reg x;
wire a, clk, rst;

// specify state machine
always @ (posedge clk)
    if (rst == 1'b1) begin
        state <= S_0;
        x <= 1'b0;
    end
else
    case (state)
        S_0: if (a==1'b1) begin
                state <= S_0; x <= 1'b0; end
            else begin
                state <= S_1; x <= 1'b0; end
            S_1: if (a==1'b1) begin
                state <= S_2; x <= 1'b0; end
            else begin
                state <= S_3; x <= 1'b1; end
            S_2: begin state <= S_3; x <= 1'b1; end
            S_3: if (a==1'b1) begin
                state <= S_1; x <= 1'b1; end
            else begin
                state <= S_0; x <= 1'b0; end
            default: begin state <= S_0; x <= 1'b0; end
        endcase
    endmodule

```

Figure 1.5a

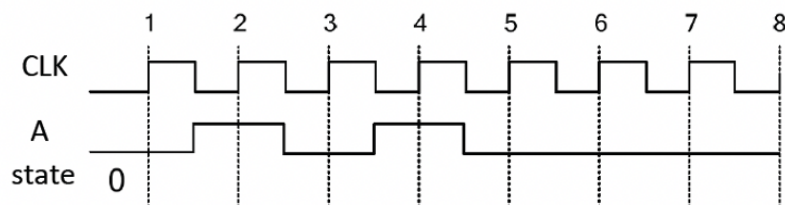


Figure 1.5b

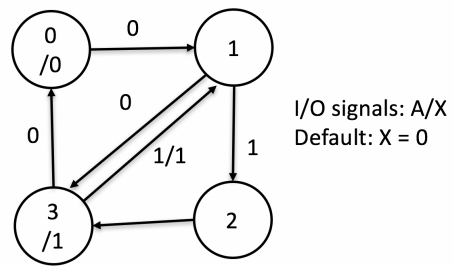
Q1 e) (i) In one-hot encoding, each state is identified by one bit in the state code.

Therefore a N-bit state machine would require N flip flops. In binary state encoding, the binary number is used to represent the state code. Therefore a N-bit state machine would only need $\lceil \log_2 N \rceil$ flip flops.

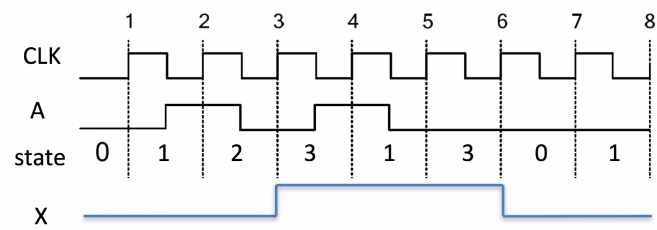
One-hot encoding is particularly suited to FPGA architecture because:

- 1) Generally it has less combinational logic because state decoding is not required. FPGA LUTs can generally only implement simple Boolean equations with few variables. This fine-grain architecture matches the simpler logic well.
- 2) FPGAs LEs contains D-FF which makes the architecture FF rich. Hence having to use more FFs in one-hot encoded state machine generally does not consume more hardware resources since the FF comes free.

(ii) The state diagram is: [4]



The timing diagram is: [4]



2. Figure 2.1 (a) shows a digital circuit that computes the approximate square root of an 8-bit number $P[7:0]$ and produces a 4-bit answer $X[3:0]$. The circuit consists of a squaring circuit, a digital comparator and a finite state machine that implements the successive approximation algorithm. The digital comparator output *HIGHER* is high if the value at its input *A* is larger than its input *B*. The FSM is controlled by the clock signal *CLK*. It starts the computation when *LOAD* goes from low to high and the *DONE* signal then goes low until the computation is completed and $X[3:0]$ contains the valid result.

- (a) Assuming that $P[7:0] = 8'h54$, complete the timing diagram shown in figure 2.1 (b).

[7]

- (b) Hence or otherwise, explain how this circuit is able to computer the square root of the input. State any assumptions used.

[8]

- (c) Design in Verilog HDL the finite-state machine SAR FSM.

[10]

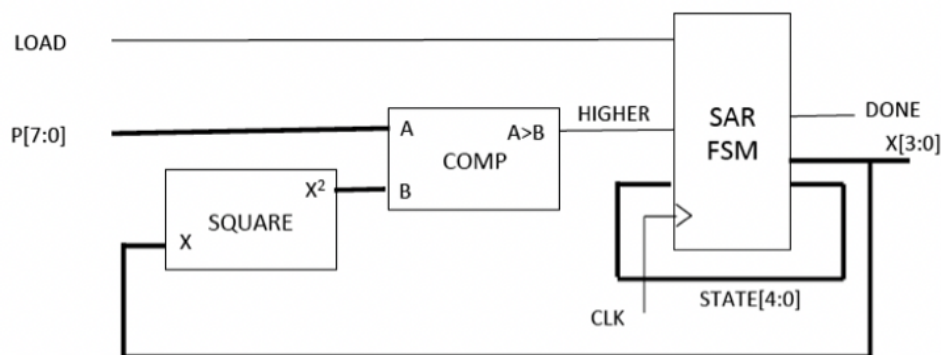


Figure 2.1 (a)

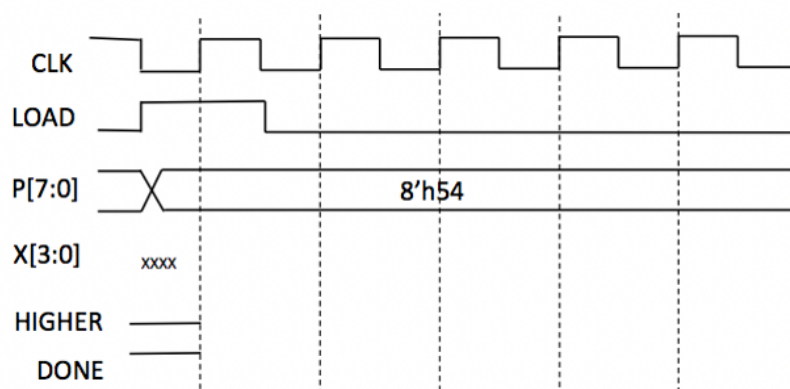


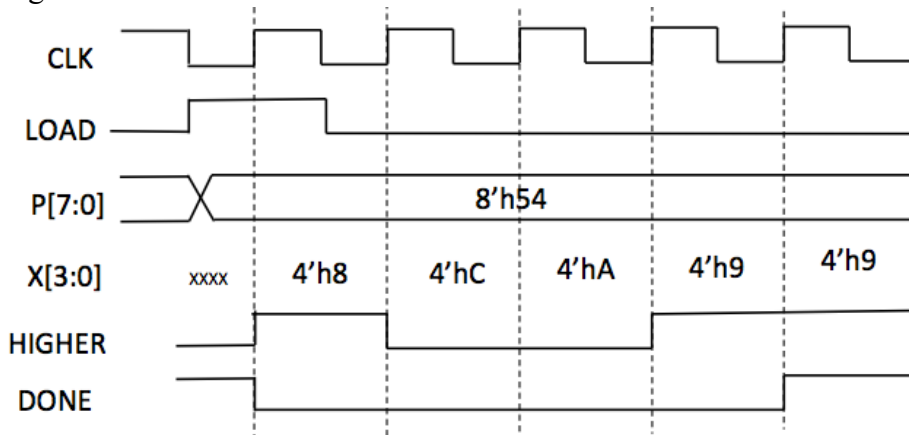
Figure 2.1 (b)

- Q2. This question tests student's understanding the principle of successive approximation algorithm and how to apply it..

- (a) Students may first draw this table to help, but not required.

$X_{[3:0]}$	X^2	HIGHER	RESULT
4h8	8h40	TRUE	$X_{[3]} = 1$
4hC	8h90	FALSE	$X_{[2]} = 0$
4hA	8h64	FALSE	$X_{[1]} = 0$
4h9	8h51	TRUE	$X_{[0]} = 1$

Timing diagram:



[7]

- (b) The principle of SAR is really bookwork. However, the application here is rather special. Instead of the DAC in a SAR based ADC, we use the squaring circuit. The rest of the SAR algorithm is exactly the same. Assumption is that LOAD is asserted just before CLK going high, and P[7:0] is stable.

[8]

- (c) The solution depends on student's code. The most obvious solution is to implement the FSM as a 31 state machine as shown in Lec 10 slide 14. One could use few states by adding extra logic inside the FSM to set and reset the bits etc., but that becomes more complex than required for a 4-bit SAR FSM.

[10]

3. *Figure 3.1* shows a circuit consisting of a 2-bit up counter CTR2, a 3-bit counter CTR3 which counts up when EN is high, a 2-input AND gate, a 3-bit register FF which is enabled when G2 is high and a 3-bit binary subtractor SUB. The circuit is clocked by the signal CLK, and has an input A and an output Z[2:0].

(a) Specify this circuit in Verilog HDL.

[10]

(b) Assuming that both counters and the register are initially reset, complete the timing diagram in *figure 3.2* by showing the values of K[1:0], Q[2:0], Y[2:0] and Z[2:0].

[15]

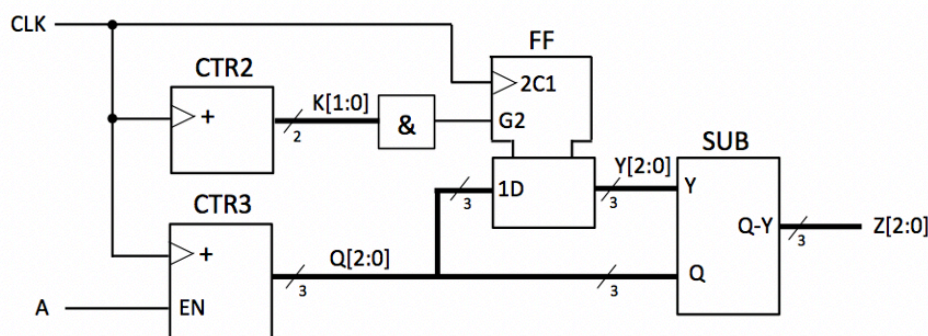


Figure 3.1

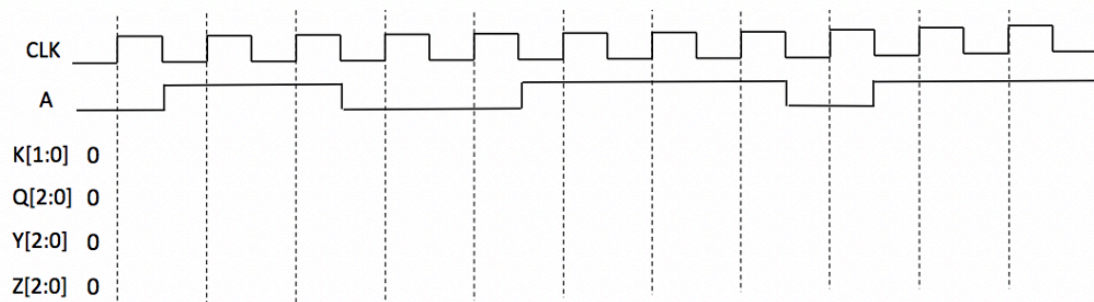


Figure 3.2

Q3. This question tests student's ability to map a reasonably complex digital circuit to Verilog, and able to predict the behaviour of the circuit.

(a)

```
module fred (clk, a, z);

    input          clk;
    input          a;
    output reg [2:0] z;

    reg [1:0]      k;
    reg [2:0]      q, y;

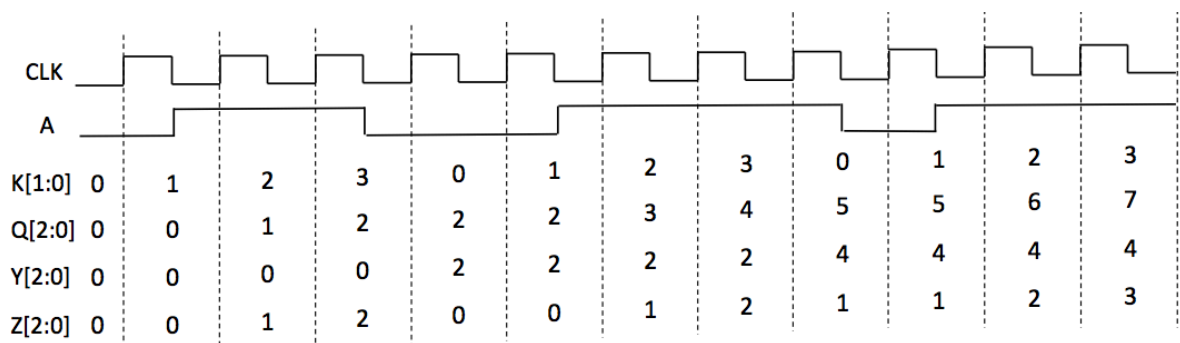
    initial k = 2'b0;
    initial q = 3'b0;
    initial y = 3'b0;

    always @ (posedge clk) begin
        if (k == 2'b11) y = q;
        k <= k + 2'b1;
        if (a == 1'b1) q <= q + 1'b1;
    end // always

    always @ (*)
        z = q - y;
endmodule
```

[10]

(b)



[15]